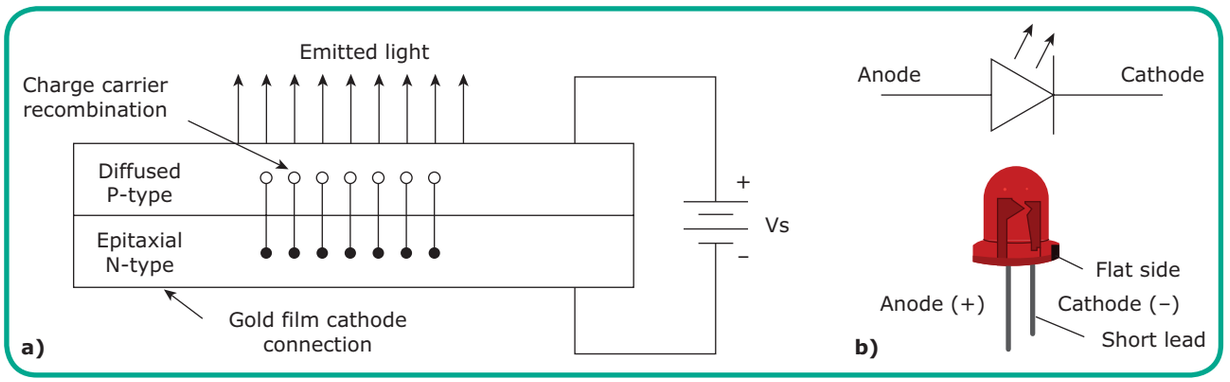


Figure 6.1 **Construction and Symbol of LED**

LEDs radiate different colours like red, green, yellow, orange, blue and white depend on material used. Some LEDs emit infrared (invisible) light also. The colour of the emitted light depends on the type of material used as given below.

- Gallium arsenide (*GaAs*) –Infrared Radiation (Invisible)
- Gallium phosphide (*GaP*) –Red or Green
- Gallium arsenide phosphide (*GaAsP*)-Red or Yellow

Working Principle

When an LED is forward biased, the electrons and holes move towards the junction and recombination takes place. This makes the electrons lying in the conduction band of *N* region fall into the holes lying in the valence band of *P* region.

The difference of energy between the conduction band and the valance band is radiated in the form of the light energy. The brightness of the emitted light is directly proportional to the forward bias current.

Advantages

- LED can be switched ON and OFF at very fast speed (1 ns).
- Its operating voltage is from 1.5 V to 3.3 V with low current (some mA).
- It has a life time of 1,00,000 hours.
- LEDs are small in size and light in weight.
- They require no heating and warm up time.

6.2 SEVEN SEGMENT LED

LEDs are often grouped to form seven segment displays. They are generally used as numerical indicators and consist of a number of LEDs arranged as seven segments as shown in the Figure 6.2.

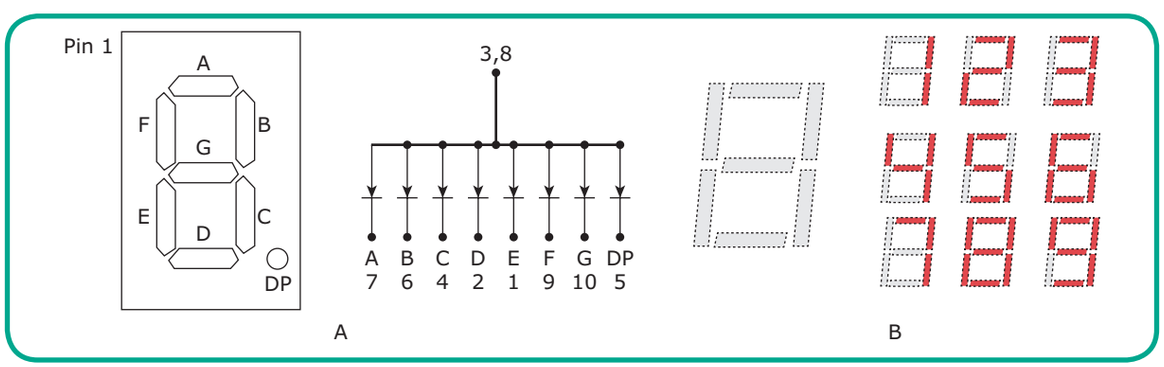


Figure 6.2 **Seven Segment LED**

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